Huawei Access Node YEL-ACC-22 QoS Congestion Event SOP

Node: YEL-ACC-22 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3809

Description

The failure of QoS congestion event (microbursts, interface tail drops, QoS queue starvation) on Huawei Access Node YEL-ACC-22 at Yelahanka Sub-station is primarily attributed to the insidious interplay of various factors inherent to the Huawei Access Node's architecture and functionality. Specifically, the occurrence of microbursts, a type of network congestion that results from the sudden and transient increase in data transmission rates, is often precipitated by the inefficient handling of bursty traffic by the Access Node's queue management algorithm. Furthermore, the inadequate configuration of Quality of Service (QoS) parameters, such as the Weighted Round Robin (WRR) algorithm, can lead to queue starvation, where one or more queues consistently receive a disproportionately low allocation of bandwidth resources, thereby exacerbating congestion. Additionally, the Huawei Access Node's reliance on the egress queueing mechanism, which is based on a First-In-First-Out (FIFO) principle, can result in interface tail drops, where packets are discarded due to the lack of available buffer space. Consequently, the confluence of these factors can lead to a QoS congestion event, which manifests as a critical alarm (ALARM_3809).

Pre-requisite Checks

- Verify the physical condition of the Huawei Access Node YEL-ACC-22 at Yelahanka Sub-station.

- Confirm the operational status of the redundant link to ensure that it is functioning correctly.

- Dispatch remote hands to Yelahanka Sub-station to perform any necessary physical checks or maintenance tasks.

Potential Root Causes

Cause	Probability
Inefficient queue management algorithm handling bursty traffic	High
Inadequate QoS parameter configuration (WRR algorithm)	Medium
Egress queueing mechanism based on FIFO principle leading to interface tail drops	Low

Diagnostic Commands & Expected Outputs

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Command: display queue statistics, Expected Output: Queue statistics for interface GigabitEthernet0/0/0: Queue 0: packets 1000000, bytes 1000000000, drops 10000 Queue 1: packets 2000000, bytes 2000000000, drops 20000, Description: This command is used to collect information about the queue statistics for the specified interface, including the number of packets and bytes transmitted, and the number of drops. This information can be used to identify any potential issues with the
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queue management algorithm or the egress queueing mechanism.

Command: display qos policy information, Expected Output: QoS policy information for interface GigabitEthernet0/0/0: Policy 0: Weight 100, Bandwidth 1000000000 Policy 1: Weight 200, Bandwidth 2000000000, Description: This command is used to retrieve information about the QoS policy configuration for the specified interface, including the weights and bandwidth allocations for each policy. This information can be used to verify that the QoS parameters are correctly configured.
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Resolution Steps

Step 1: Step: Adjust WRR algorithm weights, Command: qos policy 0 weight 150, Description: This step involves adjusting the weights of the WRR algorithm to ensure that each queue receives a more equitable allocation of bandwidth resources. The exact command and parameters will depend on the specific requirements of the network.

Step 2: Step: Configure egress queueing mechanism to avoid FIFO principle, Command: queue-profile 0 mode random-detect, Description: This step involves configuring the egress queueing mechanism to avoid the FIFO principle, which can lead to interface tail drops. Instead, the queueing mechanism is configured to use a random-detect algorithm, which can help to mitigate congestion and reduce packet drops.

Post-Check Verification

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Command: display queue statistics, Expected Output: Queue statistics for interface GigabitEthernet0/0/0: Queue 0: packets 500000, bytes 500000000, drops 500 Queue 1: packets 500000, bytes 500000000, drops 500, Description: This command is used to verify that the adjustments to the WRR algorithm and egress queueing mechanism have been successful, and that the queues are now receiving a more equitable allocation of bandwidth resources.

Command: display qos policy information, Expected Output: QoS policy information for interface GigabitEthernet0/0/0: Policy 0: Weight 150, Bandwidth 1500000000 Policy 1: Weight 150, Bandwidth 1500000000, Description: This command is used to verify that the QoS policy configuration has been updated correctly, and that the weights and bandwidth allocations for each policy are now consistent with the requirements of the network.
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Roll-back Procedures

- Step: Restore original WRR algorithm weights, Command: qos policy 0 weight 100, Description: This step involves restoring the original weights of the WRR algorithm to their previous values. This is necessary to ensure that the network returns to its previous state in the event that the adjustments made during the resolution process are not successful.

- Step: Restore original egress queueing mechanism configuration, Command: queue-profile 0 mode fifo, Description: This step involves restoring the original egress queueing mechanism configuration, which is based on the FIFO principle. This is necessary to ensure that the network returns to its previous state in the event that the adjustments made during the resolution process are not successful.